

Shuoyu Yan

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Research Interests

My current research focuses on geometric group theory, K -theory and non-commutative geometry.

Education

Ph.D. student , Université Paris Cité, IMJ-PRG	Sept 2025 – Now
• Supervised by Prof. Andrzej Zuk.	
M.Sc. in Mathematics , Université Paris-Saclay	Sept 2024 – June 2025
• Supervised by Prof. Andrzej Zuk.	
M.Sc. in Mathematics , Fudan University	Sept 2021 – June 2024
• Supervised by Prof. Yijun Yao, graduated with honours.	
B.Sc. in Mathematics , Harbin Institute of Technology	Sept 2017 – June 2021
• Supervised by Prof. Jinsong Wu.	
• Thesis: <i>Bi-shifts of bi-projections on quantum groups</i>	

Research Experience

Memoir , Université Paris Cité & IMJ-PRG	Mar 2025 – Jun 2025
<i>Spectral Analysis of Symmetric Random Walks on Groups</i> , supervised by Prof. Andrzej Zuk.	
• Summerised Kesten's work about the characterisation of amenability for groups via symmetric random walks.	
• Generalised a weaker version of Kesten-type criterion for amenable semigroups.	
• Established a characterisation of free semigroups via local and global cogrowth of a finitely generated semigroup.	
Master Thesis , Fudan University.	Nov 2022 – May 2024
<i>Global Witten deformation on Lie groupoids</i> , supervised by Prof. Yijun Yao.	
• Reviewed the classical approach of the Witten deformation on smooth manifolds.	
• Established a global approach of the Witten deformation defined by a Morse function via the DNC and blowup construction of Lie groupoids.	
• Generalised to the Morse–Bott case in particular conditions, solved the Morse inequality in the generalised case.	

Attended Events

The China-Japan-Korea Conference on Functional Analysis	Tokyo, Japan.	29/06/2026 – 03/07/2026
Quantum Theory and Operator Theory (<i>satellite conference of ICBS 2023</i>)	Beijing, China.	10/07/2023 – 14/07/2023
Special Week on Operator Algebra 2023	Shanghai, China.	31/07/2023 – 04/08/2023
YMC*A 2023	Leuven, Belgium.	07/08/2023 – 11/08/2023
Young Scholars in Noncommutative Geometry and Operator Algebras	Shanghai, China.	21/10/2023 – 22/10/2023

Teaching Experience

Teaching Assistant , Department of Mathematical Science, Fudan University	Sept 2021 – Jan 2024
• Functional Analysis, 2023 Fall.	
• Complex Analysis, 2022 Spring.	
• Principle of Mathematical Analysis, 2021 Fall.	

Languages

- **Chinese Mandarin**, mother tongue.
- **English**, Proficient, C1, *ILETS 7.0*.
- **Japanese**, Basic, A2, *JLPT N3*.